Reading Book

Jojo borrowed the book "C How to Program" in the library. Jojo wants to open selection topic on the page X. The book has N pages. Jojo can open the book from the front (from page 1) or back (from page N). Jojo will flip one sheet at a time to go to the page X. The book starts on page 1 and ends on page N. On the first sheet, the front side is page 1 and the back side is the next page. What is the minimum number of flip that must be done by Jojo to reach the page X.



Example, if $N = 4$ and $X = 3$, then 1 flip is needed (from page 1 to page 2 and 3 or from page 4 to page 2 and 3)

Format Input

The first line contains an integer T stating the number of test cases. The next T lines contains 2 integers N and X which indicate the number of pages and page numbers that Jojo wants to open.

Format Output

Consists of T lines where each line has the format "Case #X: Y", where X is the test case number starting at 1 and Y is the answer to each test case.

Constraints

- $1 \leq T \leq 10^3$
- $1 \leq N \leq 10^9$
- $1 \leq X \leq N$

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Sample Input (standard input)

```
10
100 99
100 100
100 98
100 1
100 3
5 1
5 2
5 3
5 4
5 5
```

Sample Output (standard output)

```
Case #1: 1
Case #2: 0
Case #3: 1
Case #4: 0
Case #5: 1
Case #6: 0
Case #7: 1
Case #8: 1
Case #9: 0
Case #10: 0
```

Note

In the third test case, because the book consists of 100 pages, the front of the last sheet is page 99 and the back is page 100. To go to page 98 from page 1 needs 49 page reversals (flip) and if start from page 100, only 1 page reversal (flip) is needed (from page 100 to page 99 and 98).

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Reading Book

Jojo meminjam buku "C How to Program" di perpustakaan. Jojo ingin membuka materi percabangan pada halaman X. Buku tersebut memiliki banyak halaman N. Jojo dapat membuka buku tersebut dari depan(dari halaman 1) atau pun belakang(dari halaman ke-N). Jojo akan membalik satu per satu lembar hingga menuju halaman X. Buku tersebut di mulai dari halaman 1 dan berakhir di halaman N. Pada lembar pertama, sisi depan merupakan halaman 1 dan sisi belakang adalah halaman selanjutnya. Berapa minimum pembalikan lembar yang harus dilakukan Jojo sehingga mencapai halaman ke X.



Contohnya, jika $N = 4$ dan $X = 3$, maka diperlukan 1 kali pembalikan lembar (dari halaman 1 ke halaman 2 dan 3 atau dari halaman 4 ke halaman 2 dan 3)

Format Input

Baris pertama berisi sebuah bilangan bulat T yang menyatakan banyaknya kasus uji. T baris berikutnya berisi 2 buah bilangan bulat N dan X yang menyatakan banyak halaman dan nomor halaman yang ingin dibuka Jojo.

Format Output

Terdiri dari T baris yang setiap barisnya memiliki format "Case #X: Y", dimana X adalah nomor kasus uji mulai dari 1 dan Y adalah jawaban untuk setiap kasus uji.

Constraints

- $1 \leq T \leq 10^3$

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- $1 \leq N \leq 10^9$
- $1 \leq X \leq N$

Sample Input (standard input)

```
10
100 99
100 100
100 98
100 1
100 3
5 1
5 2
5 3
5 4
5 5
```

Sample Output (standard output)

```
Case #1: 1
Case #2: 0
Case #3: 1
Case #4: 0
Case #5: 1
Case #6: 0
Case #7: 1
Case #8: 1
Case #9: 0
Case #10: 0
```

Note

Pada kasus uji ketiga, karena bukunya terdiri dari 100 halaman maka lembar terakhir bagian depannya adalah halaman 99 dan bagian belakangnya adalah halaman 100. Untuk menuju halaman 98 dari halaman 1 diperlukan 49 kali pembalikan lembar dan jika dari halaman 100 maka hanya diperlukan 1 kali pembalikan lembar (dari halaman 100 ke halaman 99 dan 98).

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